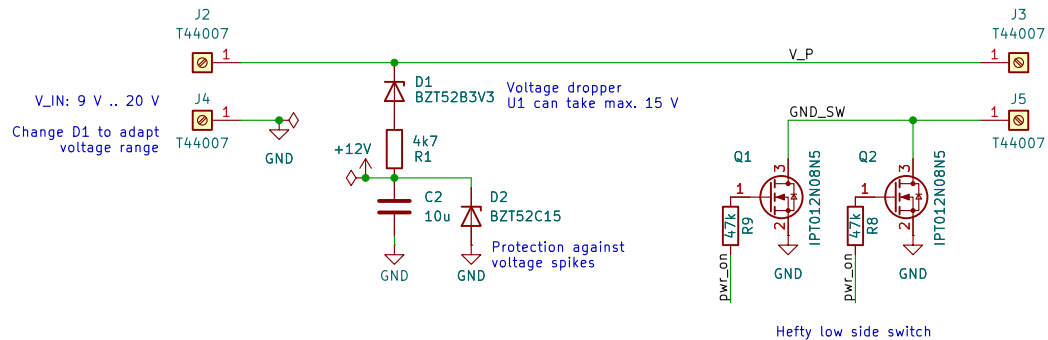
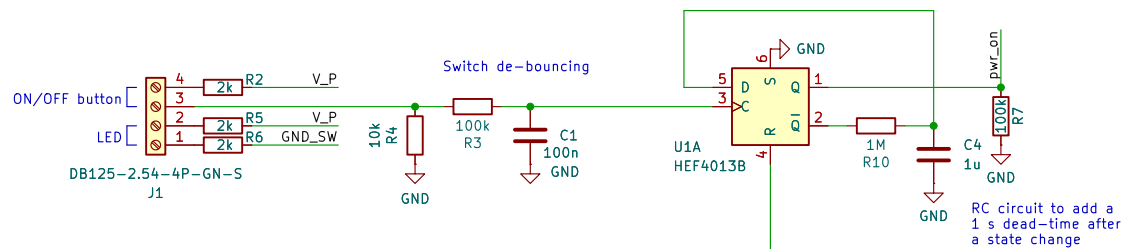




Connector options

MOLEX 389690002 (50 A)
KF78C-13.0-4P (2x 40 A)
XFCN T44007 (60 A)



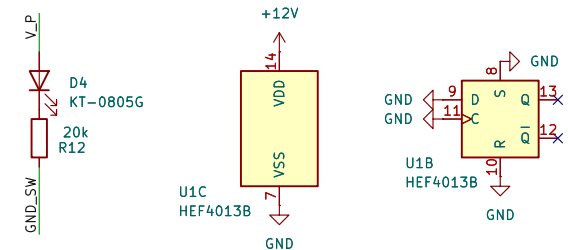
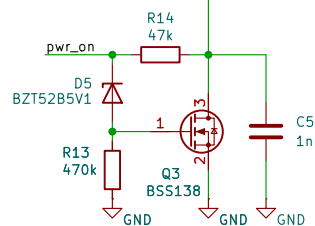
Initial power-on reset and 6 V UVLO

If pwr_on sags below 6 V, for example due to an output short, the MOSFETs Q1 and Q2 will de-saturate and get hot. But Q3 also stops conducting and RST is pulled high by R14, switching the power OFF

When power is OFF, C5 will discharge through R14, releasing the RST pin, which will allow the button to switch the power ON again. There is 0 uA standby current in OFF state.

When the power is switched ON, pwr_on rises within < 1 us, biasing Q3 before C5 can charge above the RST threshold before Q3 starts conducting, which will switch to OFF state.

During initial power-on, if the FF is in ON state, pwr_on will rise slowly (due to R1 and C2), so C5 will be charged above the RST threshold before Q3 starts conducting, which will switch to OFF state.



Rev: 1 Added UVLO circuit to protect against de-saturating the mosfets



Sheet: /
File: power_switch.kicad_sch

Title: Electronic power switch

Size: A4 Date: 2026-08-02

KiCad E.D.A. 10.0.5-rc1

Rev: 1

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